ABDALRAHMAN MUSA MOHAMED MUSA

Surveying Engineering | GIS | Remote Sensing | Data Analytics

PROFILE SUMMARY

Final-year Surveying Engineering student at the University of Khartoum with First-Class academic standing and a multidisciplinary background in surveying, GIS, remote sensing, data analytics, and Python programming. Experienced in industrial operations, materials management, data processing, and technical project development. Completed professional training from Google, NASA, and DeepLearning.AI, with practical experience using ArcGIS Pro, QGIS, Excel, and Python. Detail-oriented, adaptable, and motivated to apply technical and data-driven approaches to engineering and geospatial challenges.

CORE SKILLS

- Surveying Engineering and Geospatial Data
- GIS and Spatial Analysis
- Remote Sensing and Earth Science Applications
- Data Analytics and Data Management
- Python Programming and Automation
- Technical Problem Solving
- Inventory and Materials Management
- Quality Control and Data Accuracy

CAREER OBJECTIVE

To obtain an internship or entry-level opportunity in surveying, GIS, remote sensing, data analytics, or a related engineering field where I can apply my academic knowledge, technical skills, and practical experience while continuing to develop professionally and contribute to meaningful projects.

PERSONAL PROFILE

Location: Khartoum, Sudan

Phone: +249 99 983 2918 | +249 96 024 9184

Email: abdalrhmanuofk@gmail.com

LinkedIn: linkedin.com/in/abdalrhman-musa-372216249

EDUCATION

B.Sc. in Surveying Engineering

University of Khartoum — Faculty of Engineering

Expected Graduation: 2027

- Final-year student maintaining First-Class academic standing.

- Relevant areas include Surveying, GIS, Remote Sensing, Geodesy, Spatial Data Analysis, and Engineering Applications.

CAREER SUMMARY

Quality Control Officer

EL SHARGA Factory For Iron Production| 2019–2021

- Conducted quality-control activities during the production process to ensure materials and finished products met required specifications.

- Monitored production processes and identified quality issues to support consistent manufacturing standards.

- Coordinated with production teams to maintain workflow efficiency and product quality.

- Assisted in monitoring material movement and ensuring that processed materials were transferred to the appropriate production areas.

Strip Production Officer / Strip Processing Supervisor

EL SHARGA Factory For Iron Production

- Responsible for managing the processing of steel coils after slitting and ensuring that the resulting strips were produced according to specified widths.

- Developed and applied mathematical formulas to calculate the required strip widths based on the coil dimensions and production requirements.

- Supervised the distribution and movement of processed strips after slitting, ensuring that each batch reached the correct location.

- Coordinated the allocation of processed strips to the appropriate production machines according to operational requirements.

- Monitored the workflow after the slitting process and supervised material movement to maintain an organized production cycle.

- Worked closely with production teams to ensure accurate strip dimensions and efficient material utilization.

Raw Materials Officer

Egyptian Iron Products Factory | 2022

- Monitored stock levels for more than 200 material items.

- Maintained accurate inventory records using Microsoft Excel.

- Coordinated with suppliers and production teams to ensure timely material availability.

- Supported daily production operations and quality-control procedures.

- Trained three new employees on workplace safety procedures.

Raw Materials Department Manager

Zahrat Al-Rawdatain for Metal Factory | 2023

- Managed procurement, storage, and quality control of raw materials used in iron production.

- Implemented an inventory tracking system that contributed to reducing material waste by approximately 15%.

- Supervised a team of five warehouse staff and coordinated daily material-handling activities.

- Monitored inventory levels and supported efficient material allocation.

- Coordinated with suppliers and production teams to maintain smooth operational workflows.

Data Entry Clerk

Zain Telecommunications | 2024

- Processed more than 100 customer records daily while maintaining data accuracy.

- Verified customer information used in billing systems.

- Assisted with database maintenance and digital record organization.

- Identified and corrected data inconsistencies when required.

- Maintained confidentiality when handling customer information.

COMPUTER SKILLS

GIS & Geospatial:

ArcGIS Pro, QGIS, Spatial Analysis, Cartographic Mapping, Remote Sensing, Spatial Data Processing

Programming:

Python, C++, SQL

Data Analytics:

Microsoft Excel, Google Sheets, Tableau, Data Analysis, Data Management

Engineering & Design:

AutoCAD (Basic), CNC Systems, Photoshop

LANGUAGES

Arabic: Native

English: Professional Working Proficiency

STRENGTHS

- Strong attention to detail and commitment to accuracy
- Analytical approach to technical and data-related problems
- Quick learner of new technical systems and software
- Ability to work independently and meet deadlines
- Teamwork, coordination, and leadership
- Organized and responsible approach to tasks
- Strong interest in continuous technical and professional development

CERTIFICATIONS & PROFESSIONAL TRAINING

NASA — National Aeronautics and Space Administration

- Developing Sustainable Earth Science Applications — Module 3: An Experienced Researcher's Discovery | August 2026
- Developing Sustainable Earth Science Applications — Module 2 | August 2026
- Developing Sustainable Earth Science Applications — Module 1 | August 2026
- Fundamentals of Remote Sensing | August 2026
- Hyperspectral Data for Land and Coastal Systems | August 2026

Google

- Google Data Analytics Professional Certificate | July 2024

DeepLearning.AI

- Supervised Machine Learning: Regression and Classification | October 2024

Additional Training

- Python Programming Certification
- GIS Software Specialist
- CNC Systems Training — Mechanical Engineering Expo
- Hult Prize Competitor

GIS & TECHNICAL PROJECTS

Soil Classification Map of Khartoum Locality

GIS & Cartographic Mapping

- Developed a thematic soil classification map for Khartoum Locality using GIS and ISRIC SoilGrids data.
- Processed, classified, symbolized, and visualized spatial soil information.
- Designed a clear cartographic layout including classification, legend, labels, and essential map elements.

GIS-Based Spatial Analysis of Roads Near the Nile River

ArcGIS Pro

- Conducted spatial analysis of roads located near the Nile River.
- Used OpenStreetMap-based data and spatial datasets for roads and river features.
- Created a 5 km buffer around the Nile River to identify nearby road networks.
- Applied Buffer and Clip geoprocessing tools to perform the analysis.
- Produced a thematic map presenting the spatial relationship between roads and the river.

SurveyPy — Surveying & Geodesy Python Project

Python

- Developed a Python-based tool to automate common surveying calculations and reduce repetitive manual work.
- Implemented Bowditch traverse adjustment for distributing linear misclosures.
- Added coordinate and area calculations using the Shoelace formula.

- Developed functionality for exporting adjusted coordinates to CAD-compatible DXF files.
- Generated automated reports containing traverse misclosures and adjusted coordinates.

VOLUNTEER EXPERIENCE

Bar wa Ihsan Charity

- Participated in organizing food distribution activities supporting more than 200 families.
- Assisted with coordination and organization of charitable activities.

VMC Program

- Tutored students from underprivileged backgrounds in STEM subjects.
- Supported students in understanding technical and scientific concepts.